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Computer Science & Engineering Department

Tutorial 3:

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1. Explain the need for an I/O Module in a computer system. Describe its three primary functions (e.g., control and timing, data buffering, error detection) and list the key components required to execute these functions.
2. The CPU needs to check the status of a "Door Open" sensor in a manufacturing system. The sensor is connected via a simple I/O module. Explain the sequence of operations for this status check.
3. "A computer system uses a temperature sensor that provides new readings once per second." Which I/O processing method (Programmed I/O or Interrupt-Driven I/O) would be more appropriate for reading the sensor data? Justify your choice.
4. How does the CPU's involvement differ between Programmed I/O and Interrupt-Driven I/O?
5. Draw a block diagram showing CPU, memory, DMA controller, and an I/O device, and explain the data flow.
6. Explain with an example how bus arbitration resolves conflicts when multiple devices request the bus simultaneously.
7. Explain the process of designing a functional 8-bit ALU.
8. Explain how an ALU can perform increment, decrement, and complement operations using a 4-bit adder.
9. Describe about arithmetic, logic and shift micro operations, listing the available operations in each category.